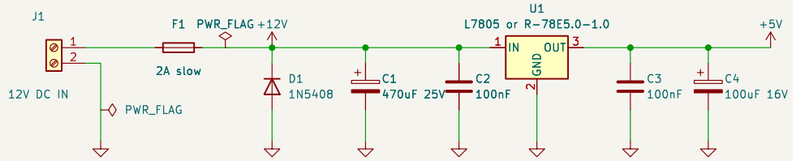
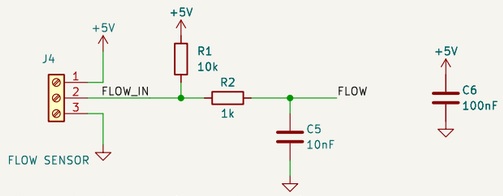
POWER: 12 V IN, FUSE, REVERSE-POLARITY CROWBAR, 5 V REGULATOR



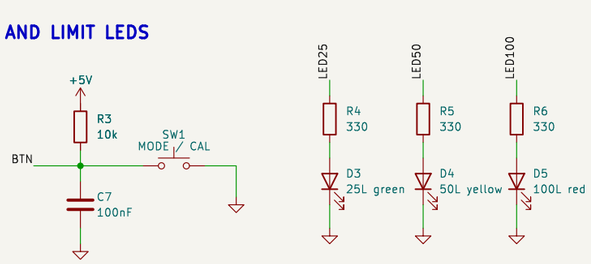
F1 + D1: if the supply is connected backwards, D1 conducts and blows F1.
U1: L7805 needs a small heatsink. Once an ESP is fitted, use a pin-compatible switching regulator (e.g. R-78E5.0-1.0) or a 5 V buck module wired to IN / GND / OUT.

FLOW SENSOR INPUT (YF-G1, NPN open collector)

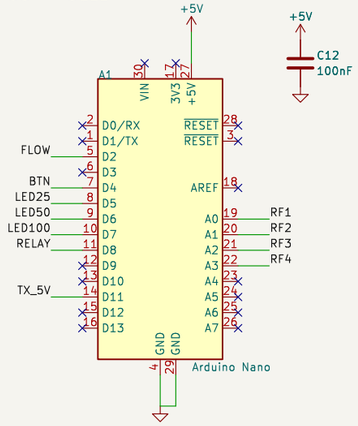


R2 + C5 filter fast spikes (time constant 10 us); the firmware also ignores pulses < 2 ms apart.
C6 sits right at J4.

BUTTON AND LIMIT LEDS

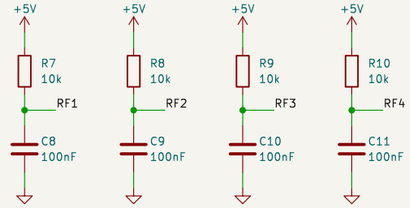


CONTROLLER

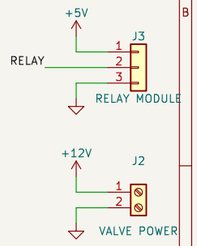


D12 is reserved for a future ESP -> Nano link.
Nano is powered through its 5V pin (VIN unused).

REMOTE RECEIVER INPUTS (433 MHz, 4 relay channels)

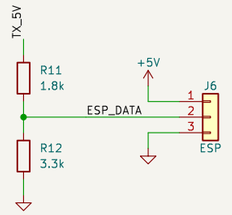


RELAY MODULE AND VALVE POWER



J2: run the valve +12 V and return as their own pair, and keep them away from the sensor and remote cables.

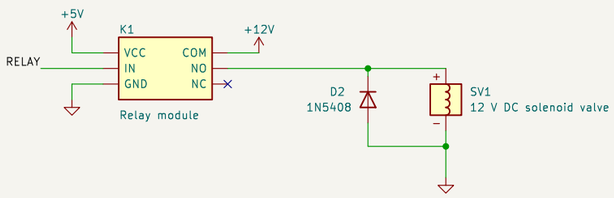
ESP DATA OUT (9600 baud, 3.3 V level)



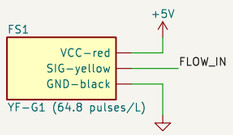
INTERFERENCE AND RELIABILITY MEASURES

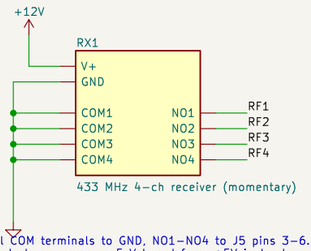
1. D2 flyback diode right at the valve terminals (stripe to +).
2. Valve current has its own wire pair (J2) back to J1: star ground at J1.
3. Decoupling: C1 + C2 on 12 V, C3 + C4 at U1, C12 at the Nano, C6 at J4.
4. Flow input: R1 10k pull-up, R2 1k + C5 10nF filter, firmware glitch filter.
5. Button and remote inputs: 10k pull-ups + 100nF, firmware debounce.
6. F1 fuse + D1 crowbar protect against shorts and a reversed supply.
7. R11 / R12 divide the 5 V data line down to 3.2 V for the ESP.
8. Twist each sensor / remote signal with its GND; route away from valve wires.
9. Ground plane on the PCB bottom layer; keep the 12 V valve tracks short and wide.

OFF-BOARD WIRING (connects to J2-J5; these parts are not placed on the PCB)



Valve circuit: relay COM <- J2 pin 1 (+12 V), valve - -> J2 pin 2.
D2 sits on the valve terminals.





Remote: all COM terminals to GND, NO1-NO4 to J5 pins 3-6.
+12 V board shown; power a 5 V board from +5V instead.

Pins: D2 flow, D4 button, D5-D7 LEDs, D8 relay, D11 ESP, A0-A3 remote
Firmware: WaterValveController.ino
YF-G1 flow sensor, relay-switched 12 V DC solenoid valve, 433 MHz remote, ESP data out
Arduino Nano carrier board

Sheet: /
File: WaterValveController.kicad_sch

Title: Water Valve Controller

Size: A3	Date: 2026-10-10	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1